EXPERIMENT 24

Title: Write a Prolog Program to Suggest Dieting System Based on Disease

Aim

To develop a Prolog program that suggests a suitable diet plan based on the disease or health condition specified by the user.

Objective

- To understand the use of facts and rules in Prolog.
- To represent disease and diet relationships.
- To implement a simple rule-based decision system.
- To provide diet suggestions based on a given condition.
- To understand the application of Prolog in expert systems.

Theory

Prolog can be used to develop simple expert systems using facts and rules. In this experiment, different diseases and their corresponding general diet recommendations are stored as facts.

When the user enters a disease as a query, Prolog searches the knowledge base and provides the corresponding diet recommendation.

This is a basic educational expert system and does not replace professional medical or dietary advice.

Algorithm

Step 1: Start the program.

Step 2: Define facts representing diseases and their recommended diets.

Step 3: Define the diet/2 predicate.

Step 4: Accept the disease as a query.

Step 5: Search the database for the corresponding diet.

Step 6: Display the recommended diet.

Step 7: If no matching disease is found, display an appropriate message.

Step 8: Stop the program.

Flowchart

```
START
  |
```

```
          ▼
   Enter Disease Name
          |
          ▼
   Search Diet Database
          |
          ▼
   Is Disease Found?
      /        \
    Yes         No
     |           |
     ▼           ▼
 Display Suitable   Display
  Diet Suggestion   "No Diet Found"
     |           |
     └───────────┬───────────┘
          |
          ▼
         STOP
```

Prolog Program

```prolog
% Disease and Diet Database

diet(diabetes, 'Low sugar and high fiber diet').
diet(hypertension, 'Low salt and low fat diet').
diet(anemia, 'Iron rich diet with green vegetables and pulses').
diet(obesity, 'Low calorie diet with fruits and vegetables').
diet(acidity, 'Light diet avoiding spicy and oily foods').
diet(constipation, 'High fiber diet with plenty of water').
```

```
% Rule to suggest diet

suggest_diet(Disease, Diet) :-
    diet(Disease, Diet).
```

Queries

Query 1: Find the diet for diabetes

```
?- suggest_diet(diabetes, Diet).
```

Sample Output

Diet = 'Low sugar and high fiber diet'.

Query 2: Find the diet for hypertension

```
?- suggest_diet(hypertension, Diet).
```

Sample Output

Diet = 'Low salt and low fat diet'.

Query 3: Find the diet for anemia

```
?- suggest_diet(anemia, Diet).
```

Sample Output

Diet = 'Iron rich diet with green vegetables and pulses'.

Query 4: Display diets for all diseases

```
?- diet(Disease, Diet).
```

Sample Output

```
Disease = diabetes,
Diet = 'Low sugar and high fiber diet' ;

Disease = hypertension,
Diet = 'Low salt and low fat diet' ;

Disease = anemia,
Diet = 'Iron rich diet with green vegetables and pulses' ;
```

Disease = obesity,

Diet = 'Low calorie diet with fruits and vegetables' ;

Disease = acidity,

Diet = 'Light diet avoiding spicy and oily foods' ;

Disease = constipation,

Diet = 'High fiber diet with plenty of water' ;

false.

Explanation

The predicate:

diet(Disease, Diet).

stores the relationship between a disease and a general diet recommendation.

For example:

diet(diabetes, 'Low sugar and high fiber diet').

represents a general dietary recommendation associated with diabetes.

The rule:

```
suggest_diet(Disease, Diet) :-
   diet(Disease, Diet).
```

is used to retrieve the diet corresponding to the disease entered in the query.

For example, when the query is:

?- suggest_diet(obesity, Diet).

Prolog searches the database and returns:

Diet = 'Low calorie diet with fruits and vegetables'.

Advantages

- Simple implementation of a rule-based expert system.
- Easy to add new diseases and diet recommendations.
- Demonstrates knowledge representation in Prolog.
- Provides quick responses to queries.

- Helps understand decision-making using logical rules.

Applications

- Basic health information systems.
- Expert systems.
- Diet recommendation systems.
- Knowledge-based Artificial Intelligence systems.
- Rule-based decision support systems.
- Educational applications.

Result

The Prolog program was successfully implemented to suggest a general diet recommendation based on the disease entered by the user. The required queries produced the corresponding results.

Conclusion

Thus, a simple disease-based diet recommendation system was successfully implemented using Prolog. This experiment demonstrates how facts and rules can be used to develop a basic expert system for decision support. The recommendations are for educational purposes and should not be considered a substitute for professional medical advice.

```
            "Vitamin C rich fruits"
        ],
        "avoid": [
            "Excess tea",
            "Excess coffee",
            "Highly processed foods"
        ]
    }
}


# Function to suggest diet
def suggest_diet(disease):
    disease = disease.lower().strip()

    if disease in diet:
        print("\nDisease:", disease.capitalize())

        print("\nRecommended Foods:")
        for food in diet[disease]["recommended"]:
            print("-", food)

        print("\nFoods to Avoid:")
        for food in diet[disease]["avoid"]:
            print("-", food)

    else:
        print("\nDisease not found in the database.")


# Main program
while True:
    print("\n--- DIET SUGGESTION SYSTEM ---")
    print("1. Get Diet Suggestion")
    print("2. Display Available Diseases")
    print("3. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        disease = input("Enter disease: ")
        suggest_diet(disease)

    elif choice == "2":
        print("\nAvailable Diseases:")
        for disease in diet:
            print("-", disease.capitalize())

    elif choice == "3":
        print("Program terminated.")
        break

    else:
        print("Invalid choice.")
```

Problems Output Debug Console Terminal Ports

```python
# DIET SUGGESTION SYSTEM BASED ON DISEASE

diet = {
    "diabetes": {
        "recommended": [
            "Green vegetables",
            "Whole grains",
            "Salads",
            "Nuts",
            "Low-sugar fruits"
        ],
        "avoid": [
            "Sugary drinks",
            "Sweets",
            "Candy",
            "Refined sugar"
        ]
    },

    "hypertension": {
        "recommended": [
            "Fresh fruits",
            "Green vegetables",
            "Whole grains",
            "Low-fat dairy",
            "Foods rich in potassium"
        ],
        "avoid": [
            "Excess salt",
            "Pickles",
            "Processed foods",
            "Fast food"
        ]
    },

    "obesity": {
        "recommended": [
            "Vegetables",
            "Fruits",
            "Whole grains",
            "Lean protein",
            "Low-fat foods"
        ],
        "avoid": [
            "Junk food",
            "Fried foods",
            "Sugary drinks",
            "High-calorie snacks"
        ]
    },

    "anemia": {
        "recommended": [
            "Spinach",
            "Beans",
            "Lentils",
            "Iron-rich foods",
```

```
--- DIET SUGGESTION SYSTEM ---
1. Get Diet Suggestion
2. Display Available Diseases
3. Exit
Enter your choice: 1
Enter disease: diabetes

Disease: Diabetes

Recommended Foods:
- Green vegetables
- Whole grains
- Salads
- Nuts
- Low-sugar fruits

Foods to Avoid:
- Sugary drinks
- Sweets
- Candy
- Refined sugar
```